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United States Patent	12402903
Kind Code	B2
Date of Patent	September 02, 2025
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Medical retrieval systems, devices, and related methods

Abstract

In one example, a retrieval device may include a flexible loop; a net coupled to the loop, the net having an opening at least partially circumscribed by the loop; a sheath extending from a proximal end to a distal end; a cord extending through and/or around the net; and a connector assembly movable from a first configuration to a second configuration. In the first configuration, the flexible loop, the net, and the cord may be coupled to the sheath, and in the second configuration, the flexible loop, the net, and the cord may be detached from the sheath.

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Appl. No.: 18/658053

Filed: May 08, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240285297 A1	Aug. 29, 2024

Related U.S. Application Data

continuation parent-doc US 17480961 20210921 US 12004762 child-doc US 18658053
us-provisional-application US 63084112 20200928

Publication Classification

Int. Cl.: A61B17/221 (20060101); **A61B17/00** (20060101)

U.S. Cl.:

CPC **A61B17/221** (20130101); A61B2017/00367 (20130101); A61B2017/2212 (20130101)

Field of Classification Search

CPC: A61B (17/221); A61B (2017/00367); A61B (2017/2212); A61B (2017/00287); A61B (2017/00358); A61B (2017/00473); A61B (2017/00477); A61B (17/00234); A61B (2018/1407); A61B (2018/141); A61B (18/1492); A61B (2017/00557); A61B (17/32056); A61B (2017/00867); A61B (17/22031); A61B (10/04)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5647372	12/1996	Tovey	600/562	A61B 17/00234
5741271	12/1997	Nakao	604/523	A61B 18/24
5997547	12/1998	Nakao et al.	N/A	N/A
8206401	12/2011	Nakao	N/A	N/A
8652147	12/2013	Hart	N/A	N/A
8906036	12/2013	Farascioni	N/A	N/A
9370341	12/2015	Ceniccola et al.	N/A	N/A
11484329	12/2021	Ahluwalia et al.	N/A	N/A
11504103	12/2021	Johnson et al.	N/A	N/A
2005/0267492	12/2004	Poncet	606/114	A61B 17/221
2006/0052799	12/2005	Middleman	606/114	A61B 17/320016
2008/0221587	12/2007	Schwartz	N/A	N/A
2009/0082780	12/2008	Lu et al.	N/A	N/A
2012/0046667	12/2011	Cherry et al.	N/A	N/A
2013/0190773	12/2012	Carlson	606/114	A61B 17/00234
2019/0336117	12/2018	Prior	N/A	A61B 17/00234
2020/0305866	12/2019	Knapp	N/A	A61B 17/00234

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2016172679	12/2015	WO	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/480,961, filed on Sep. 21, 2021, which claims the benefit of priority

TECHNICAL FIELD

(1) This disclosure generally relates to medical systems, devices, and related methods that may be used to treat a subject. Aspects of the disclosure relate to medical systems, devices, and methods for endoscopic medical procedures, such as grasping and removing foreign objects from body lumens, organs, or cavities, among other aspects.

BACKGROUND

(2) Organ walls are composed of several layers: the mucosa (the surface layer), the submucosa, the muscularis (muscle layer), and the serosa (connective tissue layer). In gastrointestinal, colonic, and esophageal cancer, lesions or cancerous masses may form along the mucosa and often extend into the lumens of the organs. Polypectomy of colonic polyps has been shown to reduce the risk of colon cancer development. A variety of polypectomy techniques and devices are available, and their use can vary greatly based on local availability and physician preferences. These procedures are generally performed with an endoscope, which is a long, narrow member optionally equipped with a light, imaging equipment, and other instruments. During these procedures, the endoscope may be passed through a percutaneous incision, passed down the throat, or guided through the rectum to reach tissue targeted for resection or dissection. Snare devices are often used in the removal of the targeted tissue, such as polyps, from a patient.

(3) During snaring, the snare may be closed around a polyp to detach the polyp from the patient's body, for example from the colon wall. The loose polyp may then be retrieved by the physician using a net or other device. The physician may insert a net through a working channel of an endoscope and deploy the net around the polyp. Then, the physician may remove the entire endoscope along with the net from the patient's body. During this process, the combination of the polyp and the net may be pulled up to the distal face of the endoscope and the view from a camera at the distal face may be obstructed. When the view of the endoscope is obstructed during withdrawal, the physician may miss cancerous tissue due to the obstructed view. Also, the working channel of the endoscope is occupied by the net retrieval device, and thus cannot be used for other purposes during the withdrawal of the endoscope from the body. If additional polyps are identified, the physician will have to re-insert the endoscope into the patient's body and re-locate the identified polyp for removal, which adds procedural time and additional stress to the patient.

SUMMARY

(4) Examples of the disclosure relate to, among other things, systems, devices, and methods for performing one or more medical procedures with the medical systems and devices. Each of the examples disclosed herein may include one or more of the features described in connection with any of the other disclosed examples.

(5) In one example, a retrieval device may include a flexible loop; a mesh coupled to the loop, the mesh having an opening at least partially circumscribed by the loop; a sheath extending from a proximal end to a distal end; a cord extending through and/or around the mesh; and a connector assembly movable from a first configuration to a second configuration. In the first configuration, the flexible loop, the mesh, and the cord may be coupled to the sheath, and in the second configuration, the flexible loop, the mesh, and the cord may be detached from the sheath.

(6) In other aspects, the retrieval device may include one or more of the following features. The cord may include a slip knot. The retrieval device may also include a drive member coupled to the flexible loop, and the actuation of the drive member may be configured to move the flexible loop relative to the sheath. The drive member may extend through the sheath, and the cord may extend outside of the sheath. The retrieval device may also include a handle including an actuator moveable in a proximal or distal direction relative to a body of the handle, and the actuator may be coupled to the drive member. The application of a proximal pulling force onto the drive member

may be configured to move the connector assembly from the first configuration to the second configuration. The cord may extend externally from the sheath. The cord may include a free end extending away from the mesh. Upon application of a pulling force on the cord, the cord may be configured to constrict at least a portion of the mesh. Upon constriction of the mesh, the mesh may form a substantially-enclosed bag or sac. The mesh may be configured to remain constricted when the flexible loop, the mesh, and the cord are decoupled from the sheath in the second configuration of the connector assembly. The cord may be woven into the mesh. The cord may be not directly coupled to the flexible loop. The flexible loop may be a snare loop. The connector assembly may include a capsule having a lumen and the flexible loop includes proximal arms that extend into the lumen; and the capsule may be configured to separate from the sheath when the connector assembly moves from the first configuration to the second configuration.

(7) In other aspects, a retrieval device may include a snare loop; a mesh coupled to the snare loop, the mesh having an opening at least partially defined by the snare loop; a sheath extending from a proximal end toward a distal end; a cord extending through and/or around the mesh, and upon application of a pulling force on the cord, the cord is configured to constrict at least a portion of the mesh; and a drive member coupled to the snare loop. The actuation of the drive member may be configured to move the snare loop relative to the sheath.

(8) In other aspects, the retrieval device may include one or more of the following features. The snare loop, the mesh, and the cord may be configured to be decoupled from the sheath. Application of a pulling force on the drive member may be configured to decouple the snare loop, the mesh, and the cord from the sheath.

(9) In other aspects, a method of retrieving an object from a body may include inserting a medical device with a working channel into the body; extending a sheath, and a snare loop and a mesh coupled to the sheath through the working channel and distally of the medical device; placing the mesh around the object; constricting at least a portion of the mesh around the object; and decoupling the snare loop, the mesh, and the object from the sheath. In some examples, the method may further include removing the snare loop, the mesh, and the object from the body separately from the medical device.

(10) It may be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of this disclosure, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate exemplary aspects of the disclosure and together with the description, serve to explain the principles of the disclosure.

(2) FIG. 1 illustrates a perspective view of a medical device system, according to aspects of this disclosure.

(3) FIG. 2 illustrates a perspective view of an exemplary medical device, according to aspects of this disclosure.

(4) FIG. 3A illustrates a perspective view of a distal portion of the exemplary medical device in FIG. 2, according to aspects of this disclosure.

(5) FIG. 3B illustrates a perspective view of a distal portion of the exemplary medical device in FIG. 2, according to aspects of this disclosure.

(6) FIGS. 4-5 illustrate a distal end of a medical device with portions shown in cross-section, according to aspects of this disclosure.

DETAILED DESCRIPTION

(7) The terms “proximal” and “distal” are used herein to refer to the relative positions of the components of an exemplary medical system and exemplary medical devices. When used herein, “proximal” refers to a position relatively closer to the exterior of the body or closer to a medical professional using the medical system or medical device. In contrast, “distal” refers to a position relatively further away from the medical professional using the medical system or medical device, or closer to the interior of the body. Proximal and distal directions are labeled with arrows marked “P” and “D”, respectively, throughout the figures. As used herein, the terms “comprises,” “comprising,” “having,” “including,” or other variations thereof, are intended to cover a non-exclusive inclusion, such that a system, device, or method that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent thereto. Unless stated otherwise, the term “exemplary” is used in the sense of “example” rather than “ideal.” As used herein, the terms “about,” “substantially,” and “approximately,” indicate a range of values within $\pm 10\%$ of a stated value.

(8) Embodiments of this disclosure include devices, systems, and methods for manipulating, cutting, collecting, retrieving, and/or otherwise carrying tissue. In some examples the devices, systems and/or methods discussed herein may be utilized during endoscopic mucosal resection (EMR), endoscopic submucosal dissection (ESD), polypectomy, and/or other medical procedures. In examples, polypectomy includes endoluminal placement of a retrieval device and a cutting device proximate to a target area within the body of a patient. Placement of the retrieval and cutting devices may be via a catheter, scope (endoscope, bronchoscope, colonoscope, gastroscope, duodenoscope, etc.), tube, or sheath, inserted into the GI tract via a natural orifice or incision. The orifice can be, for example, the nose, mouth, or anus, and the placement can be in any portion of the GI tract, including the esophagus, stomach, duodenum, large intestine, or small intestine. Placement also can be in other organs reachable via the GI tract. A polyp within the body of the patient may be cut by the cutting device for subsequent removal from the patient's body. The retrieval device may collect the polyp and move the polyp out of the patient's body.

(9) Reference will now be made in detail to examples of this disclosure described above and illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

(10) FIG. 1 illustrates a perspective view of an exemplary medical device assembly **100** including an endoscope **101** and a medical device **200**. Although medical device assembly **100** is shown with endoscope **101**, any other similar insertion device may be used in medical device assembly **100**, such as a bronchoscope, colonoscope, gastroscope, duodenoscope, etc. Endoscope **101** may include a handle **102**, actuators **104**, and a body **107** extending from handle **102** to a distal end **108**. A working channel **106** may extend from a working channel port **103** positioned on the handle **102** to an opening at distal end **108**. Distal end **108** of endoscope **101** may also include a camera **105**, and movement of distal end **108** and functionality of camera **105** may be controlled via one or more actuators **104** on handle **102**. Medical device **200** may be configured to be inserted into working channel **106** of endoscope **101**.

(11) Medical device **200** may include a handle **202**, a main sheath or shaft **204** extending from the handle **202** to a distal portion **206** of medical device **200**, and a snare assembly **208** at the distal portion **206** of the device. Main shaft **204** may extend from handle **202** into working channel **106** of endoscope **101**, and may be movable within working channel **106**. Main shaft **204** may extend longitudinally for a length that is larger than the longitudinal length of endoscope **101**, such that once main shaft **204** is pushed through working channel **106**, a portion of main shaft **204** extends distally from a distal opening of working channel **106** (shown in FIG. 1). Snare assembly **208** at distal portion **206** of medical device **200** may include a flexible snare loop **210**, a net **212**, a connector assembly **214**, and a cord **220**. Cord **220** may be coupled to snare loop **210** and net **212**. Cord **220** may extend longitudinally by a length larger than the longitudinal length of endoscope **101**, such that once cord **220** is moved through working channel **106**, a portion of cord **220** extends

distally from a distal opening of working channel **106**, while a portion of cord **220** extends proximally out of working channel port **103** (shown in FIG. **1**).

(12) FIG. **2** illustrates a perspective view of medical device **200** from FIG. **1** removed from endoscope **101**. Handle **202** may include an actuator **249** including two loops (or holders) **250**, **251** configured to receive fingers of an operator of the medical device, and handle **202** may also include a third loop (or holder) **252** at a proximal end of handle **202**. Loops **250**, **251**, and **252** may be rings, grips, or other similar structures that include an aperture to receive one or more fingers of a user. Actuator **249** may be moveably coupled to body **224** of handle **202**, and may be configured to slide in distal and proximal directions to move a drive wire **418** (shown in FIG. **4**). Drive wire **418** may extend longitudinally from handle **202** through shaft **204** to connector assembly **214**, and drive wire **418** may be configured to move snare loop **210** into and out of shaft **204** and/or connector assembly **214**. Snare loop **210** is shown deployed in FIG. **2**, and may be configured to be fully received within connector assembly **214** and/or shaft **204**, or may be configured to be partially received within connector assembly **214** and/or shaft **204** when actuator **249** is pulled proximally by an operator. By moving actuator **249** longitudinally (distally or proximally), an operator may adjust the deployed size of snare loop **210**. Cord **220** may extend from distal portion **206** to handle **202** and the positioning of cord **220** may allow an operator to pull cord **220** proximally during operation.

(13) Cord **220** may be coupled to net **212** and may include a knot (e.g., a slip knot) **302** (FIG. **3A**) tied at a proximal portion of snare loop **210** (when snare loop **210** is in the deployed, ready-to-capture configuration shown in FIG. **3A**). In some examples, cord **220** may be coupled to net **212** by weaving cord **220** through holes of net **212**, glued or otherwise adhered to net **212**, and/or fastened to net **212** via one or more fasteners (e.g., stapled or clipped). FIG. **3A** shows an exemplary distal portion **206** of medical device **200** with cord **220** weaved through net **212** to couple cord **220** to net **212**. Cord **220** may be a string, wire, suture, line, or any other suitable rope known in the art. In some examples, slip knot **302** may extend around a proximal portion of snare loop **210** (not shown). In other examples, slip knot **302** is positioned proximate to net **212** and does not extend around a proximal portion of snare loop **210**, such that slip knot **302** is positioned spaced apart from (above or below) snare loop **210** (shown in FIG. **3A**). Slip knot **302** may be configured to allow an operator to pull cord **220** proximally and configured to prevent separation of cord **220** from net **212**. For example, when an operator pulls cord **220** proximally, cord **220** may gather net **212** into a bundle and form an enclosed (or partially-enclosed) pouch with net **212**, securing any captured items or material (e.g., a polyp, stone, or other object) within the pouch. Slip knot **302** may allow cord **220** to be pulled proximally to bring portions of net **212** together and prevent tissue or other material gathered within net **212** from moving out of net **212**. Although slip knot **302** is shown in FIG. **3A**, other fastening mechanisms may be used to prevent separation of cord **220** from net **212** while allowing proximal movement of cord **220**, such as a spring biased connector or other fastening mechanism known in the art.

(14) Net **212** may be fixedly coupled to snare loop **210** via glue, fasteners, or other adhesives/mechanisms known in the art. In some examples, net **212** may be coupled to snare loop **210** via coating or enveloping snare loop **210** with a bonding material, and then coupling a plurality of strands of net **212** to the bonding material. In some examples, portions of net **212** are embedded in the bonding material. Cord **220** may extend circumferentially around net **212**, may be coupled to portions of net **212** proximate to snare loop **210**, and may form a loop portion **300**. By coupling cord **220** to net **212** proximate to snare loop **210**, a sufficient amount of net **212** may be positioned on an opposing side of net **212** from snare loop **210** (either above or below cord **220**) and available to collect tissue. Tissue may be transported in net **212** when cord **220** is pulled proximally to collect portions of net **212** together and form a bag or pouch assembly shown in FIG. **3B**.

(15) Connector assembly **214** may provide a mechanism to couple drive wire **408** to snare loop **210**. Connector assembly **214** may be cylindrical and may be longitudinally aligned with shaft **204**.

In some examples, connector assembly **214** may be integral with shaft **204**. Connector assembly **214** may be configured to provide a mechanism for the operator to adjust the size of the deployed snare loop **210** (e.g. change the circumference of the snare loop **210** via actuator **249**), and provide a mechanism to detach snare loop **210**, net **212**, and cord **220** from shaft **204**. In some examples, connector assembly **214** may be configured to decouple from shaft **204** when a user pulls drive wire **408** proximally with at least a threshold level of force. In other examples, handle **202** may include an actuator (separate from actuator **249**) and connector assembly **214** may be configured to decouple from shaft **204** when the actuator is actuated (e.g., a button is pressed or a lever is pulled). Connector assembly **214** will be discussed in further detail below in relation to FIGS. **4** and **5**.

(16) FIG. **3B** shows snare loop **210** after a user has pulled cord **220** to tighten portions of net **212** together. As shown in FIG. **3B**, cord **220** enables a user to pull cord **220** to bunch or sync portions of net **212** together and form a bag with net **212** around tissue (or object) **350**. Snare loop **210** is decoupled from shaft **204** in FIG. **3B**, and may be pulled proximally via cord **220** without moving shaft **204**. Slip knot **302** is positioned away from (above or below) snare loop **210** proximate to net **212**. Since loop **300** of cord **220** is coupled to portions of net **212** (e.g., weaved through holes of net **212**), portions of net **212** are brought together when a user pulls cord **220** proximally.

(17) In operation of medical device system **100**, a user may first insert endoscope **101** into a body of a patient and position distal portion **108** proximate to a target area within the patient's body. A user may visualize the target area using a camera **105** of endoscope **101**. The user may then insert distal portion **206** of medical device **200** through working channel **106** until distal portion **116** exits a distal opening of working channel **106**. In other examples, a user may position medical device **200** within working channel **106** prior to insertion of endoscope **101** into the body of the patient. Once distal portion **206** exits working channel **106** and is positioned proximate to the target area, a user may move actuator **249** proximally or distally relative to body **224** of handle **202** to open and close snare loop **210**. The user may then position net **212** around tissue and at least partially close snare loop **210** by pulling actuator **249** proximally. Once snare loop **210** is closed, the user may then pull cord **220** proximally to bring portions of net **212** together to form a bag with net **212** holding the tissue. The user may then pull actuator **249** further proximally to release snare loop **210** from shaft **204** via connector assembly **214** decoupling from shaft **204**. Once snare loop **210** is released, the user may remove snare loop **210** from the body of the patient by pulling cord **220** proximally, and then may continue operating on patient using endoscope **101**. Once treatment of the patient is completed, the user may fully remove medical device system **100** from the patient's body.

(18) FIG. **4** illustrates a distal portion **406** of a medical device **401** including snare loop **410**, cord **420**, connector assembly **414**, and drive wire **418**. Medical device **401**, and any of its components, may include any of the features described herein in connection with medical device **101**. FIG. **4** shows a more detailed view of an exemplary decoupling mechanism generally referred to in the previous figures. Medical device **401** may include a handle assembly identical to handle **202**, and specifically medical device **401** may include an actuator **249**. Connector assembly **414** is disposed at the distal portion of medical device **401**, and contains a mechanism that converts the proximal and distal movement of control wire **418** into the actions necessary to actuate (e.g. expand/contract) and release snare loop **410**. Some of the components of the connector assembly **414** include a capsule **400** which provides a structural shell for the connector assembly **414**, snare loop **410** which moves between expanded and contracted positions, a bushing **422** attached to the distal end of the control wire **418**, and a yoke **404** adapted to connect the capsule **400** to the control wire **418**.

(19) As depicted in FIG. **4**, the proximal end of the capsule **400** slides over the distal end of the bushing **422**. A locking arrangement between these two components is provided by tabs on capsule **400**, which lock into bushing **422** so that mechanical integrity is temporarily maintained between the capsule **400** and the bushing **422**. Within capsule **400** are a yoke **404** and a tension member **406** which transmit forces applied by the control wire **418** to snare loop **410**. A ball **440** formed at the

distal end of the control wire **418** is mated to a receiving socket formed at the proximal end of the yoke **404**. In some examples, tension member **406** is received in yoke **404**, so that the two components are releasably connected to one another. Snare loop **410**, while in a closed or collapsed configuration, may be partially contained within the capsule **400** to prevent outward expansion of snare loop **410** from a central longitudinal axis **475** of device **401**. Each proximal arm **481**, **482** of snare loop **410** may extend through and be coupled to the tension member **406**, and may have a proximal end which slips under a yoke overhang, to further control movement of the snare loop **410**.

(20) FIG. 4 shows a view of medical device **401** with snare loop **410** in an open configuration. The open configuration is obtained when ball **440** of the control wire **418** pushes the assembly containing the yoke **404** and the tension member **406** distally, sliding snare loop within and ultimately distal to the capsule **400**. Snare loop **410** may be materially biased toward the open position and revert to this position whenever snare loop **410** is not constrained by the capsule **400**. Snare loop **410** may include proximal arms **481**, **482** that extend within capsule **400**. In some examples, tabs **445** provide a cam surface, and the proximal arms **481**, **482** of snare loop **410** act as cam followers, being deflected by the tabs **445**. In addition, the folding tabs **445** may also provide a distal stop for the tension member **406**, to retain tension member **406** within the capsule **400**.

(21) When the actuator **249** is moved proximally by the user, the yoke **404** and tension member **406** within the capsule **400** also moves proximally and snare loop **410** is withdrawn within the capsule **400**. As proximal arms **481**, **482** move proximally within the capsule **400**, clip stop shoulders (CSS) **491**, **492** contact a distal portion of the capsule **400**, for example, the folded tabs **445**. CSS **491**, **492** may be rigid protrusions that extend radially inward towards a center of snare loop **410**, may be triangular or any other suitable shape, and may be configured to engage with tabs **445**. During contraction of snare loop **410**, the interaction of the CSS **491**, **492** with the capsule **400** provides to the user a first tactile feedback in the form of increased resistance to movement of sliding actuator **249**. This feedback gives to the operator a positive indication that further movement of the handle control will cause the snare loop **410** to be detached from the connector assembly **414**. The operator may then decide whether the current position of the snare loop **410** (e.g., within the body) is acceptable or not. If the position is acceptable, the operator can release snare loop **410** from a proximal portion of connector assembly **414** (and the body of medical device **401**) by continuing to move actuator **249** with increased proximal pressure to cause the snare loop **410** to close. If not, the operator can move actuator **249** distally to re-open snare loop **410** and extend snare loop **410** out of the capsule **400**, reposition the snare loop **410**, and repeat the above steps to close snare loop **410** at a more appropriate location.

(22) When the user determines that snare loop **410** is positioned correctly, the proximal pressure on actuator **249** may be increased to continue detachment of the snare loop **410** from connector assembly **414**. As the proximal tension on actuator **249** is increased, control wire **418** pulls the yoke **404** proximally, away from the tension member **406**. The tension member **406** is firmly attached to the proximal arms **481**, **482** which are prevented from moving proximally by the interaction of the CSS **491**, **492** with the folded tabs **445**. If sufficient pulling force is applied to the yoke **404**, tension member **406** may yield and lose integrity and/or lose connection with the yoke **404**. In other words, after a sufficient proximal force, yoke **404** will decouple from tension member **406**. This can occur because, in some examples, the tension member **406** is formed of a material with a lower yield strength than the material of the yoke **404**.

(23) The force required to decouple the tension member **406** from the yoke **404** may be tailored to achieve a desired feedback that can be perceived by the user. The minimum force required to break the tension member **406** free of the yoke **404** may be selected so that a tactile feedback is felt by the user, to prevent premature release of snare loop **410** while a maximum force may be selected so that other components of the linkage between actuator **249** and proximal arms **481**, **482** do not fail before tension member **406** and yoke **404** disconnect from one another. In one exemplary

embodiment, the tension force necessary to disconnect the two components may be in the range of about 4 lbf to about 12 lbf, although these values are not limiting. This range may vary depending on the size of the device and the specific application. To obtain this force at the interface of yoke **404** and tension member **406**, a larger force may be applied by the user at actuator **249**, since friction within the device may cause losses along the long flexible shaft.

(24) When yoke **404** decouples from tension member **406**, several events take place within medical device **401** nearly simultaneously. More specifically, the yoke **404** is no longer constrained from moving proximally by the CSS **491**, **492** abutting the capsule **400**. Thus, yoke **404** travels proximally until it comes to rest against a distal shoulder of bushing **422**. The tension member **406** is not affected by this movement since it is no longer connected to the yoke **404**.

(25) A further result of moving the yoke **404** against a distal shoulder of the bushing **422** is that the distal end of a wire stop **460** is placed near a proximal bushing shoulder of bushing **422**. Flared fingers **462** located at the distal end of the wire stop **460** are compressed as they pass through the bushing **422**, but return to their normally biased open position after moving past a proximal bushing shoulder of bushing **422**. Further distal movement of actuator **249** is thus prevented because the fingers **462** of wire stop **460** would abut against the proximal bushing shoulder of bushing **422**. This abutment prevents connector assembly **414** from being pushed away from the bushing **422** before the ball **440** is separated from the control wire **418**, as will be described below.

(26) The wire stop **460** may include a tube with a first slotted and flared end attached to the control wire **418** by conventional mechanisms. The slots imparts flexibility to the control wire **418** so it can easily pass through the central lumen of the bushing **422**. Flared fingers **462** are formed by the slots, and engage the proximal bushing shoulder. The wire stop **460** may include a material that is biocompatible and that has enough resilience so that the fingers **462** re-open after passage through the bushing **422**. For example, stainless steel may be used for this application.

(27) In at least some examples, the user receives both tactile and auditory feedback as the connector assembly **414** is deployed and released. The separation of the tension member **406** from the yoke **404** may produce a clicking noise and a tactile feel that is perceptible while holding handle **202**. The change in axial/longitudinal position of actuator **249** is thus augmented by the changes in resistance to its movement and by the clicking sound and tactile feedback through the start and stop of the movement. As a result, the user may always be aware of the status of the connector assembly **414**, and the inadvertent release of snare loop **410** in an incorrect location is less likely. It will be apparent to those of skill in the art that the order of components (e.g., male and female connectors in the device may be reversed or changed without affecting the operation of the device.

(28) FIG. 5 depicts a configuration where connector assembly **414** has separated from a remainder of the medical device **401**. The control wire **418** may be designed so that it separates from the end ball **440** only after a predetermined tension is applied to control wire **418**. In other words, the ball **440** is configured to yield and separate from the body of the control wire **418** when a tension applied to control wire **418** reaches and/or exceeds a pre-set level.

(29) As the ball **440** detaches, actuator **249** bottoms out at the proximal end of the handle **202**, such that a full stroke of the handle assembly is reached. The tension required to cause the ball **440** to release from control wire **418** may be any suitable value. However, this tension force may be greater than the tension force required for the tension member **406** to separate from the yoke **404**. If this condition is not satisfied, a situation may occur where the connector assembly **414** is locked in place, but cannot be released from medical device **401**.

(30) Once the ball **440** has separated from the remainder of the control wire **418**, the user may pull cord **420**. As this is done, the yoke **404** is retained within the capsule **400** by the spring and frictional forces of various features of the capsule **400**, such as the capsule tabs. The user may then remove snare loop **410** including yoke **404**, tension member **406**, and capsule **400**, from the patient, by pulling cord **420** proximally. Further description and additional embodiments of connector

assemblies similar to connector assembly **414** may be found in U.S. Pat. No. 8,062,311, issue on Nov. 22, 2011, which is incorporated by reference herein in its entirety. Any of the clip assemblies or aspects of clip assemblies disclosed in the '311 patent application may be used as connector assemblies or other features in any of the medical devices disclosed herein.

(31) In other examples, after decoupling of snare loop **210** from shaft **204**, the proximal end of cord **220** could be tied off or otherwise secured at the proximal end of endoscope **101** with suture clips or other locking mechanisms. This would allow the captured material to remain trapped, while also keeping the material and net **212** out of direct view of camera **105**, enabling a user to continue operating with endoscope **101**.

(32) In some examples, multiple medical devices **200** may be used with endoscope **101**, and the user may snare tissue and remove multiple snare loops **210** from a patient's body during a procedure without removing endoscope **101**. In some examples, snare loops **210** may be removed by pulling cord **220** proximally and moving snare loops **210** proximally outside of endoscope **101**, and/or snare loops **210** may be removed from patient by pulling cord **220** proximally through a working channel of endoscope **101**.

(33) It also should be understood that one or more aspects of any of the medical devices, systems, and methods described herein may be used for cutting, capturing, holding, dissecting, treating, or ablating tissue in any part of the human body. For example, any of the medical devices described herein may be used in medical procedures such as for Endoscopic Submucosal Dissection (ESD), cancer treatment, and/or other procedures where removal of the type of tissue is needed.

(34) Various aspects discussed herein may help reduce procedure time, increase tissue treatment effectiveness, reduce the risks to the subject, etc.

(35) Although the exemplary embodiments described above have been disclosed in connection with medical devices for manipulating and capturing human tissue through the working channel of a medical device, a natural orifice, or by incision, a person skilled in the art will understand that the principles set out above can be applied to any medical device or medical method and can be implemented in different ways without departing from the scope of the disclosure as defined by the claims. In particular, constructional details, including manufacturing techniques and materials, are well within the understanding of those of skill in the art and have not been set out in any detail here. These and other modifications and variations are well within the scope of the this disclosure and can be envisioned and implemented by those of skill in the art.

(36) Moreover, while specific exemplary embodiments may have been illustrated and described collectively herein, it should be appreciated that any subsequent arrangement designed to achieve the same or similar purpose may be substituted for the specific embodiments described and shown herein. This disclosure is intended to cover any and all subsequent adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, will be apparent to those of skill in the art upon reviewing the description.

(37) While principles of the disclosure are described herein with reference to illustrative aspects for particular applications, it should be understood that the disclosure is not limited thereto. Those having ordinary skill in the art and access to the teachings provided herein will recognize additional modifications, applications, aspects, and substitution of equivalents all fall within the scope of the aspects described herein. Accordingly, the disclosure is not to be considered as limited by the foregoing description.

(38) Other exemplary embodiments of the this disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the exemplary embodiments disclosed herein. It is intended that the specification and examples be considered as exemplary only, and departures in form and detail may be made without departing from the scope and spirit of the this disclosure as defined by the following claims.

Claims

1. A method of retrieving an object from a body, the method comprising: inserting a medical device with a working channel into the body; extending a sheath, an expandable wire loop, a net coupled to the expandable wire loop, and a fastening mechanism coupled to the net, through the working channel, and distally of the medical device, wherein the expandable wire loop further comprises a capsule; placing the net around the object; actuating an actuator on a handle, wherein the actuator is moveable in a proximal or a distal direction relative to a body of the handle, and wherein, after placing the net around the object, the actuator is moved proximally to at least partially close the expandable wire loop; wherein, after placing the net around the object, moving the actuator further proximally moves the expandable wire loop partially within the capsule; applying a pulling force on the fastening mechanism to close at least a portion of the net around the object; and decoupling the expandable wire loop, the net, and the object from the sheath by actuating a drive wire coupled to the expandable wire loop, wherein the drive wire is configured to move the expandable wire loop relative to the sheath.
2. The method of claim 1, wherein the net has an open end and a closed end, wherein the open end is at least partially circumscribed by the expandable wire loop, wherein the closed end is spaced apart from the open end, and wherein, prior to inserting the medical device into the body, the fastening mechanism is coupled between the open end and the closed end of the net.
3. The method of claim 1, wherein, after the expandable wire loop is partially within the capsule, continuing movement in the proximal direction produces a first feedback indicating that further proximal movement decouples the expandable wire loop from the sheath.
4. The method of claim 3, wherein continuing movement in the proximal direction after the first feedback decouples the expandable wire loop, the net, and the object from the sheath.
5. The method of claim 4, wherein decoupling the expandable wire loop from the sheath is configured to produce a second feedback indicating decoupling has occurred.
6. The method of claim 1, wherein the fastening mechanism is a cord, and wherein the pulling force on the fastening mechanism is a proximal force.
7. The method of claim 1, wherein, upon closing the net, the net forms a substantially-enclosed bag or sac.
8. The method of claim 1, further comprising: removing the expandable wire loop, the net, and the object from the body separately from the medical device.
9. A method of retrieving an object from a body, the method comprising: inserting a medical device with a working channel into the body, wherein the medical device includes a handle with an actuator; extending a sheath, a snare loop, a net coupled to the snare loop, and a cord coupled to the net, through the working channel, and distally of the medical device, wherein a drive wire is coupled to the snare loop, and wherein moving the actuator tensions the drive wire to move the snare loop relative to the sheath, and wherein the snare loop further comprises a capsule; positioning the net around the object, wherein the actuator is moveable in a proximal direction or in a distal direction relative to a body of the handle, wherein after placing the net around the object, the actuator is moved proximally to at least partially close the snare loop; wherein, after placing the net around the object, moving the actuator further proximally moves the snare loop partially within the capsule; moving the actuator to at least partially close the snare loop; applying a force on the cord to close at least a portion of the net around the object; and decoupling the snare loop, the net, and the object from the sheath by moving the actuator to reach a threshold level of force.
10. The method of claim 9, wherein after the snare loop is partially within the capsule, the actuator is configured to reach the threshold force by continuing proximal movement.
11. The method of claim 9, wherein the force applied on the cord is a proximal force.
12. The method of claim 9, wherein, upon closing the net, the net forms a substantially-enclosed

bag or sac.

13. A method of retrieving an object from a body, the method comprising: inserting a medical device with a working channel into the body, wherein the medical device includes a handle with an actuator; extending a sheath, a snare loop, wherein the snare loop further comprises a capsule, a net coupled to the snare loop, and a cord coupled to the net, through the working channel, and distally of the medical device; positioning the net around the object; moving the actuator in a proximal direction relative to a body of the handle, wherein after placing the net around the object, the actuator is moved proximally to at least partially close the snare loop, wherein, after placing the net around the object, moving the actuator further proximally moves the snare loop partially within the capsule; applying a proximal force on the cord to close at least a portion of the net around the object; and decoupling the snare loop, the net, and the object from the sheath by moving the actuator further proximally.
